

Digital Image Processing

Beginner-Friendly Theoretical Guide, Proofs, Intuitions & Worked Examples

1 Introduction to Digital Images, Sampling, and Image Topology

1.1 Intuitive Foundations of Digital Images

At its most fundamental level, a digital image is a two-dimensional grid of intensity values. Mathematically, we represent an image as a function $f(x, y)$, where (x, y) specifies the spatial coordinate (row and column position) and $f(x, y)$ denotes the brightness or intensity level at that point.

In digital systems, both the spatial coordinates (x, y) and the intensity values $f(x, y)$ must be finite, discrete quantities.

- **Spatial Resolution:** Refers to the density of pixels forming the grid (e.g., $M \times N$ pixels).
- **Intensity Resolution (Bit-Depth):** Refers to the number of discrete intensity levels L assigned to each pixel, typically expressed as $L = 2^k$, where k is the bit-depth. For an 8-bit image, $k = 8$, yielding $L = 256$ possible intensity levels ranging from 0 (pure black) to 255 (pure white).

1.2 Perceptual Foundations: Brightness Adaptation and False Contouring

The human visual system (HVS) does not perceive absolute light intensity in a linear fashion. Instead, human visual perception operates according to logarithmic brightness adaptation, governed by the **Weber-Fechner Law**:

$$\frac{\Delta I_c}{I} = K_{Weber} \approx 0.01 \text{ to } 0.02 \quad (1)$$

where I represents the background intensity and ΔI_c is the minimum change in intensity required for a human observer to perceive a difference (known as the *just-noticeable difference*).

Beginner Intuition: What Causes False Contouring?

When an image with smooth intensity gradients (such as a sunset or sky) is stored using too few discrete intensity levels (e.g., 4 bits or $L = 16$ levels), the jump between two adjacent discrete grey levels becomes larger than ΔI_c .

Because the human eye is acutely sensitive to abrupt intensity steps, these coarse quantization jumps appear as artificial, step-like boundary lines running through smooth regions. This visual artifact is called **false contouring** (or the Mach band effect).

1.3 The 2-D Nyquist-Shannon Sampling Theorem

To record a continuous real-world scene $f(x, y)$ as a digital image without losing structural detail, we must sample it at discrete spatial intervals Δx and Δy .

Mathematical Mechanism of Spatial Sampling

Sampling a continuous function $f(x, y)$ can be modeled mathematically as multiplying $f(x, y)$ by a two-dimensional array of Dirac delta functions (an impulse train or Dirac comb) $s(x, y)$:

$$s(x, y) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(x - m\Delta x, y - n\Delta y) \quad (2)$$

Multiplication in the spatial domain corresponds to convolution in the frequency domain. As a result, the continuous Fourier spectrum $F(u, v)$ of the original image is replicated periodically across the 2-D frequency plane at intervals of $\frac{1}{\Delta x}$ and $\frac{1}{\Delta y}$:

$$F_s(u, v) = \frac{1}{\Delta x \Delta y} \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} F\left(u - \frac{m}{\Delta x}, v - \frac{n}{\Delta y}\right) \quad (3)$$

To prevent these periodic spectral copies from overlapping (an effect known as **aliasing**), the sampling rates must satisfy the **2-D Nyquist-Shannon Sampling Condition**:

$$\frac{1}{\Delta x} \geq 2u_{max}, \quad \frac{1}{\Delta y} \geq 2v_{max} \quad (4)$$

where u_{max} and v_{max} represent the highest horizontal and vertical spatial frequencies present in the original continuous image. If we sample at a rate lower than $2u_{max}$, high-frequency spatial details fold back into lower frequencies, creating distorted patterns such as moiré fringes.

1.4 Pixel Neighborhoods and Topological Ambiguity Resolution

To analyze region shapes and boundaries, we must formally define how adjacent pixels relate to one another spatially.

Let p be a pixel located at coordinate (x, y) . Its basic spatial neighborhoods are defined as:

- **4-Neighbors** $N_4(p)$: The four orthogonal pixels immediately above, below, left, and right of p :

$$N_4(p) = \{(x+1, y), (x-1, y), (x, y+1), (x, y-1)\}$$

- **Diagonal Neighbors** $N_D(p)$: The four corner pixels diagonally adjacent to p :

$$N_D(p) = \{(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)\}$$

- **8-Neighbors** $N_8(p)$: The union of 4-neighbors and diagonal neighbors:

$$N_8(p) = N_4(p) \cup N_D(p)$$

Standard 8-connectivity frequently introduces topological ambiguities, such as multiple concurrent paths or self-intersecting boundaries. To solve this, we use **mixed connectivity** (m -connectivity).

Definition & Step-by-Step Proof: m -Connectivity

Two pixels p and q with intensity values belonging to a specified set V are m -connected if: 1. $q \in N_4(p)$, OR 2. $q \in N_D(p)$ AND the intersection set $N_4(p) \cap N_4(q)$ contains **no** pixels with intensity values from V .

Proof of Topological Path Uniqueness: Consider a 2×2 matrix with $V = \{1\}$:

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

Let $p = (0, 0)$ (top-left pixel) and $q = (1, 1)$ (bottom-right pixel).

1. Under Standard 8-Connectivity: Pixel p connects to q directly through the diagonal link in $N_D(p)$. Additionally, p connects to q via the 4-neighbor path $(0, 0) \rightarrow (0, 1) \rightarrow (1, 1)$ and $(0, 0) \rightarrow (1, 0) \rightarrow (1, 1)$. This yields **three simultaneous paths**, creating ambiguity regarding object boundaries.

2. Under Mixed m -Connectivity: We evaluate $N_4(p) \cap N_4(q) = \{(0, 1), (1, 0)\}$. Since both of these intermediate pixels have intensity values of $1 \in V$, the second condition of m -connectivity fails. Consequently, the diagonal connection between p and q is eliminated. Path multiplicity collapses to a single, unambiguous path sequence.

2 2-D Discrete Fourier Transform, Spectrum, and Convolution

2.1 Beginner Intuition: What is the Fourier Transform?

The Fourier Transform provides an alternative way of looking at an image. Instead of viewing an image as a spatial arrangement of pixel brightnesses $f(x, y)$, the Fourier Transform decomposes the image into a weighted sum of 2-D sinusoidal waves of varying frequencies, amplitudes, and directions.

- **Low Spatial Frequencies:** Represent smooth, gradually varying regions (e.g., broad backgrounds, soft skin tones).
- **High Spatial Frequencies:** Represent rapid intensity changes (e.g., sharp edges, fine text, noise).

2.2 Mathematical Formulation

For a digital image $f(x, y)$ of size $M \times N$, the **2-D Discrete Fourier Transform (DFT)** converts spatial intensities into frequency coefficients $F(u, v)$:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} \quad (5)$$

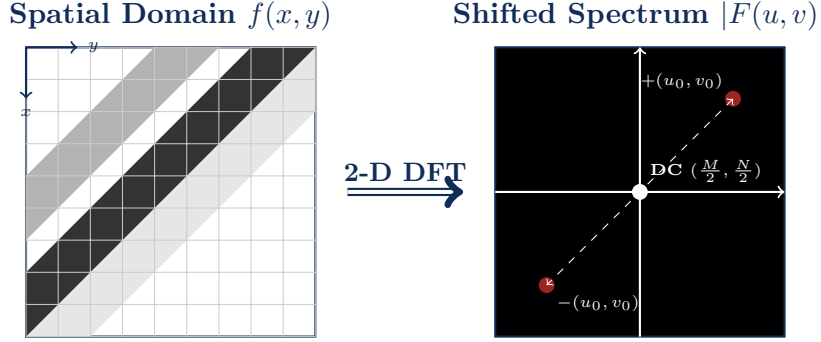
for frequency variables $u = 0, 1, \dots, M-1$ and $v = 0, 1, \dots, N-1$.

The original spatial image is reconstructed using the **2-D Inverse Discrete Fourier Transform (IDFT)**:

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})} \quad (6)$$

Textbook Motivation: Why Process Images in the Frequency Domain?

Periodic interference patterns (such as electrical hum lines superimposed on a camera sensor) spread across the entire spatial image, making them difficult to isolate. However, in the Fourier domain, periodic sinusoidal patterns map to isolated, concentrated impulse points at specific frequencies $\pm(u_0, v_0)$. By zeroing out those specific frequency points, we eliminate the noise while leaving the rest of the image intact.



2.3 Proof of 2-D DFT Separability

The 2-D DFT kernel $e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$ is separable into a product of 1-D functions: $e^{-j2\pi\frac{ux}{M}} \cdot e^{-j2\pi\frac{vy}{N}}$. This property allows us to compute a 2-D DFT by executing 1-D DFTs sequentially across rows and columns.

Proof. By expanding and factoring the 2-D sum:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi\frac{ux}{M}} e^{-j2\pi\frac{vy}{N}} \quad (7)$$

$$= \sum_{y=0}^{N-1} \left[\sum_{x=0}^{M-1} f(x, y) e^{-j2\pi\frac{ux}{M}} \right] e^{-j2\pi\frac{vy}{N}} \quad (8)$$

Let $G(u, y) = \sum_{x=0}^{M-1} f(x, y) e^{-j2\pi\frac{ux}{M}}$ be the 1-D DFT of column y evaluated at horizontal frequency u . Substituting $G(u, y)$ into the outer summation yields:

$$F(u, v) = \sum_{y=0}^{N-1} G(u, y) e^{-j2\pi\frac{vy}{N}} \quad (9)$$

which is simply the 1-D DFT along row u of $G(u, y)$.

This reduces the computational complexity from $\mathcal{O}(M^2N^2)$ operations down to $\mathcal{O}(MN \log_2(MN))$ operations using the Fast Fourier Transform (FFT) algorithm. $\square$

2.4 Proof of the 2-D Circular Convolution Theorem

The 2-D circular convolution of two padded $M \times N$ spatial functions $f(x, y)$ and $h(x, y)$ is defined as:

$$g(x, y) = f(x, y) \otimes h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h((x - m) \bmod M, (y - n) \bmod N) \quad (10)$$

Proof. Taking the 2-D DFT of the convolved signal $g(x, y)$:

$$G(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \left[\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h((x-m) \bmod M, (y-n) \bmod N) \right] e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} \quad (11)$$

Exchanging the order of summation:

$$G(u, v) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) \left[\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} h((x-m) \bmod M, (y-n) \bmod N) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} \right] \quad (12)$$

Let $x' = (x - m) \bmod M$ and $y' = (y - n) \bmod N$. Due to the periodic nature of the complex exponential kernel $e^{-j2\pi k} = 1$, as x ranges from 0 to $M - 1$, x' also traverses 0 to $M - 1$:

$$\sum_{x=0}^{M-1} e^{-j2\pi \frac{u(x'+m)}{M}} = e^{-j2\pi \frac{um}{M}} \sum_{x'=0}^{M-1} e^{-j2\pi \frac{ux'}{M}} \quad (13)$$

Substituting this relation back into the nested sum:

$$G(u, v) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) e^{-j2\pi(\frac{um}{M} + \frac{vn}{N})} \left[\sum_{x'=0}^{M-1} \sum_{y'=0}^{N-1} h(x', y') e^{-j2\pi(\frac{ux'}{M} + \frac{vy'}{N})} \right] \quad (14)$$

$$= F(u, v) \cdot H(u, v) \quad (15)$$

This proves that spatial convolution is mathematically equivalent to point-by-point multiplication in the Fourier frequency domain. $\square$

3 Histogram-Based Processing and Thresholding

3.1 Image Histograms and Continuous Equalization Proof

The normalized histogram of an L -level digital image ($L = 256$) measures the relative probability of occurrence of each intensity level r_k :

$$p_r(r_k) = \frac{n_k}{MN}, \quad k = 0, 1, \dots, L - 1 \quad (16)$$

where n_k is the number of pixels with intensity level r_k , and MN is the total pixel count.

Mathematical Derivation: Why CDF Equalizes Histograms

Let intensity levels be treated as continuous variables $r \in [0, L - 1]$ governed by Probability Density Function (PDF) $p_r(r)$. We wish to construct a monotonic mapping $s = T(r)$ such that the output intensity PDF $p_s(s)$ becomes perfectly uniform: $p_s(s) = \frac{1}{L-1}$.

Consider the Cumulative Distribution Function (CDF) transformation:

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw \quad (17)$$

Differentiating $s = T(r)$ with respect to r using the Fundamental Theorem of Calculus yields:

$$\frac{ds}{dr} = (L - 1)p_r(r) \quad (18)$$

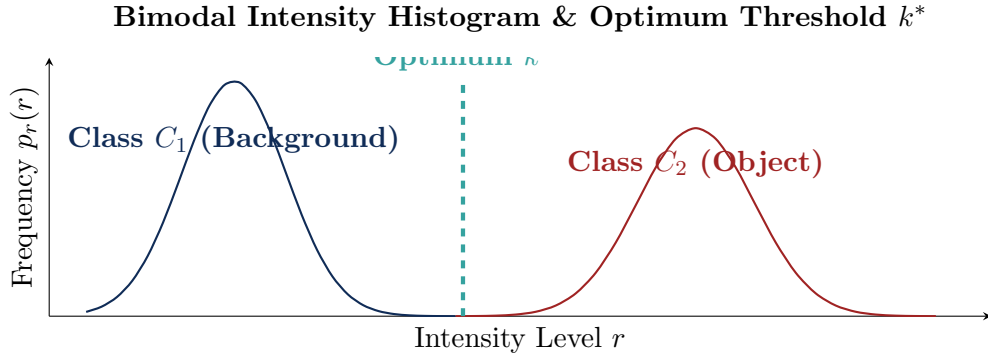
According to the fundamental transformation law for random variables:

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right| = p_r(r) \cdot \frac{1}{(L - 1)p_r(r)} = \frac{1}{L - 1}, \quad 0 \leq s \leq L - 1 \quad (19)$$

This completes the proof. Applying the CDF transformation guarantees a uniform output intensity distribution.

3.2 Otsu's Global Thresholding Derivation

Otsu's method automatically calculates the optimal intensity threshold k^* that segments an image into background pixels $C_1 = [0 \dots k]$ and object pixels $C_2 = [k + 1 \dots L - 1]$ without human intervention.



Proof of Between-Class Variance Optimization. Total variance σ_T^2 is constant and splits into within-class variance σ_W^2 and between-class variance σ_B^2 : $\sigma_T^2 = \sigma_W^2(k) + \sigma_B^2(k)$.

Minimizing within-class variance is equivalent to maximizing between-class variance:

$$\sigma_B^2(k) = P_1(k)(m_1(k) - m_G)^2 + P_2(k)(m_2(k) - m_G)^2 \quad (20)$$

where:

- $P_1(k) = \sum_{i=0}^k p_i$ is the cumulative probability of Class C_1 .
- $P_2(k) = 1 - P_1(k)$ is the cumulative probability of Class C_2 .
- $m(k) = \sum_{i=0}^k i p_i$ is the cumulative mean up to level k .

- $m_G = \sum_{i=0}^{L-1} i p_i$ is the global mean intensity of the image.

Substituting class means $m_1(k) = \frac{m(k)}{P_1(k)}$ and $m_2(k) = \frac{m_G - m(k)}{1 - P_1(k)}$ into $\sigma_B^2(k)$:

$$\sigma_B^2(k) = P_1 \left(\frac{m}{P_1} - m_G \right)^2 + (1 - P_1) \left(\frac{m_G - m}{1 - P_1} - m_G \right)^2 \quad (21)$$

$$= \frac{(m - P_1 m_G)^2}{P_1} + \frac{(P_1 m_G - m)^2}{1 - P_1} \quad (22)$$

$$= (m_G P_1(k) - m(k))^2 \left[\frac{1}{P_1(k)} + \frac{1}{1 - P_1(k)} \right] \quad (23)$$

$$= \frac{[m_G P_1(k) - m(k)]^2}{P_1(k)[1 - P_1(k)]} \quad (24)$$

Otsu's algorithm computes $\sigma_B^2(k)$ for $k = 0 \dots L - 1$ and selects $k^* = \arg \max_k \sigma_B^2(k)$. □

Solved Example: Step-by-Step Histogram Equalization Numerical Calculation

A 64×64 pixel 3-bit digital image ($L = 8$ intensity levels, 0 through 7) has the following pixel count distribution:

Intensity Level (r_k)	0	1	2	3	4	5	6	7
Pixel Count (n_k)	790	1023	850	656	329	245	122	81

Compute the histogram equalized output levels s_k .

Solution: Total pixel count $MN = 64 \times 64 = 4096$. Scale factor $\frac{L-1}{MN} = \frac{7}{4096} \approx 0.00170898$.

Using formula $s_k = \text{round} \left(\frac{7}{4096} \sum_{j=0}^k n_j \right)$:

k	n_k	Cumulative Sum $\sum n_j$	Exact Value s_k	Final Mapped Level $\text{round}(s_k)$
0	790	790	$7 \times 790/4096 = 1.350$	1
1	1023	1813	$7 \times 1813/4096 = 3.098$	3
2	850	2663	$7 \times 2663/4096 = 4.551$	5
3	656	3319	$7 \times 3319/4096 = 5.672$	6
4	329	3648	$7 \times 3648/4096 = 6.234$	6
5	245	3893	$7 \times 3893/4096 = 6.653$	7
6	122	4015	$7 \times 4015/4096 = 6.861$	7
7	81	4096	$7 \times 4096/4096 = 7.000$	7

Resulting intensity mapping: $r_0 \rightarrow 1$, $r_1 \rightarrow 3$, $r_2 \rightarrow 5$, $r_3, r_4 \rightarrow 6$, and $r_5, r_6, r_7 \rightarrow 7$.

4 Spatial Filtering, Edge Operators, and Wiener Restoration

4.1 Taylor Series Derivation of Discrete Derivatives

To construct discrete filters that detect edges, we approximate spatial derivatives by expanding $f(x + h, y)$ and $f(x - h, y)$ in Taylor series expansions about (x, y) :

$$f(x + h, y) = f(x, y) + h \frac{\partial f}{\partial x} + \frac{h^2}{2!} \frac{\partial^2 f}{\partial x^2} + \frac{h^3}{3!} \frac{\partial^3 f}{\partial x^3} + \mathcal{O}(h^4) \quad (25)$$

$$f(x - h, y) = f(x, y) - h \frac{\partial f}{\partial x} + \frac{h^2}{2!} \frac{\partial^2 f}{\partial x^2} - \frac{h^3}{3!} \frac{\partial^3 f}{\partial x^3} + \mathcal{O}(h^4) \quad (26)$$

Setting pixel step size $h = 1$:

- **First Derivative Approximation (Forward Difference):**

$$\frac{\partial f}{\partial x} \approx f(x + 1, y) - f(x, y) \quad (27)$$

- **Second Derivative Approximation (Central Difference):** Summing both Taylor expansions:

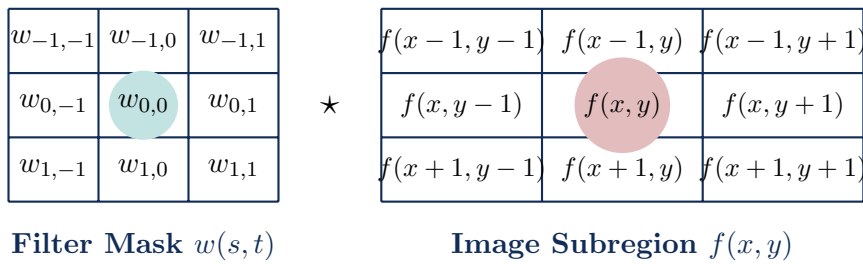
$$\frac{\partial^2 f}{\partial x^2} = f(x + 1, y) - 2f(x, y) + f(x - 1, y) \quad (28)$$

Combining partial derivatives along both horizontal and vertical axes defines the isotropic **Laplacian Operator**:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = f(x + 1, y) + f(x - 1, y) + f(x, y + 1) + f(x, y - 1) - 4f(x, y) \quad (29)$$

4.2 Spatial Mask Grid Layout

Spatial filtering operates by sliding a filter mask $w(s, t)$ across every pixel in image subregion $f(x, y)$:



4.3 Wiener Filter (Minimum Mean Square Error Restoration)

When an image is degraded by blur function $h(x, y)$ and additive noise $\eta(x, y)$, the degradation model in the frequency domain is $G(u, v) = H(u, v)F(u, v) + N(u, v)$.

Simple inverse filtering $\hat{F}(u, v) = \frac{G(u, v)}{H(u, v)} = F(u, v) + \frac{N(u, v)}{H(u, v)}$ fails because $H(u, v) \rightarrow 0$ at high frequencies causes noise $N(u, v)$ to blow up infinitely.

The **Wiener Filter** solves this problem by deriving an optimal filter function $W(u, v)$ that minimizes the

expected mean square error $E\{|f(x, y) - \hat{f}(x, y)|^2\}$:

$$W(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + \frac{S_\eta(u, v)}{S_f(u, v)}} = \frac{1}{H(u, v)} \left[\frac{|H(u, v)|^2}{|H(u, v)|^2 + K} \right] \quad (30)$$

where $S_\eta(u, v) = |N(u, v)|^2$ is noise power spectrum, $S_f(u, v) = |F(u, v)|^2$ is signal power spectrum, and $K = \frac{S_\eta}{S_f}$ is the noise-to-signal power ratio.

5 Morphological Image Processing

Morphology extracts image components useful in representing and describing region shapes (e.g., boundaries, skeletons).

5.1 Fundamental Morphological Operations

Let A represent a binary image and B represent a structuring element (SE) with origin $(0, 0)$.

Definitions of Fundamental Morphological Operators

- **Erosion** ($A \ominus B$): $A \ominus B = \{z \mid (B)_z \subseteq A\}$. A pixel z belongs to the eroded image if B translated to z fits entirely inside A . (Shrinks foreground objects).
- **Dilation** ($A \oplus B$): $A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$. A pixel z belongs to the dilated image if B translated to z touches at least one foreground pixel in A . (Expands foreground objects).
- **Opening** ($A \circ B$): $A \circ B = (A \ominus B) \oplus B$. Erosion followed by dilation. Smooths object contours, breaks thin connections, and eliminates small isolated noise spikes.
- **Closing** ($A \bullet B$): $A \bullet B = (A \oplus B) \ominus B$. Dilation followed by erosion. Smooths contours, fuses narrow gaps, and fills small holes.

Solved Example: Worked Example: Erosion vs. Dilation

Given binary matrix A and a 3×3 cross-shaped structuring element B centered at $(1, 1)$:

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Calculate Erosion $A \ominus B$ and Dilation $A \oplus B$.

Solution: 1. **Erosion** ($A \ominus B$): Testing B centered at position $(2, 2)$ (the exact middle pixel of A): The 4-neighbors of $(2, 2)$ in A are all 1s. Thus B fits completely inside A **only** at position $(2, 2)$.

$$A \ominus B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2. **Dilation** ($A \oplus B$): Any 0 pixel adjacent to a 1 in A becomes 1:

$$A \oplus B = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

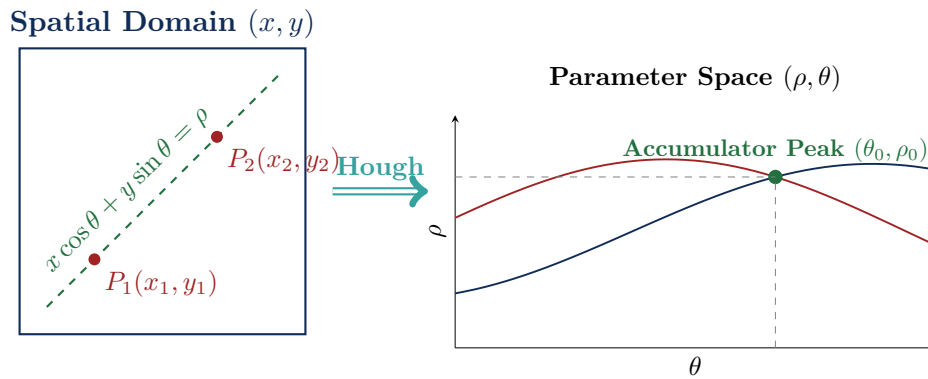
6 Advanced Segmentation and Parameter Space Transformations

6.1 Hough Transform Parameterization

In Cartesian coordinates, a line $y = ax + b$ suffers from slope singularity as lines approach vertical orientation ($a \rightarrow \infty$). To solve this, Duda and Hart introduced normal polar parameterization:

$$x \cos \theta + y \sin \theta = \rho \quad (31)$$

where ρ is the perpendicular distance from the origin to the line, and $\theta \in [-90^\circ, 90^\circ]$ is the angle of the normal vector. Collinear spatial points (x_i, y_i) transform into intersecting sinusoidal curves in the (ρ, θ) parameter space.



Solved Example: Step-by-Step Hough Parameter Intersection Problem

Given two spatial edge points $P_1(1, 1)$ and $P_2(2, 2)$, compute the (ρ, θ) accumulator values for $\theta \in \{-45^\circ, 0^\circ, 45^\circ, 90^\circ\}$ and identify the common intersecting parameter cell.

Solution: Formula: $\rho = x \cos \theta + y \sin \theta$.

1. For $P_1(1, 1)$:

- $\theta = -45^\circ \implies \rho = 1 \cos(-45^\circ) + 1 \sin(-45^\circ) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = \mathbf{0}$
- $\theta = 0^\circ \implies \rho = 1 \cos(0^\circ) + 1 \sin(0^\circ) = 1(1) + 0 = 1$
- $\theta = 45^\circ \implies \rho = 1 \cos(45^\circ) + 1 \sin(45^\circ) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2} \approx 1.414$
- $\theta = 90^\circ \implies \rho = 1 \cos(90^\circ) + 1 \sin(90^\circ) = 0 + 1(1) = 1$

2. For $P_2(2, 2)$:

- $\theta = -45^\circ \implies \rho = 2 \cos(-45^\circ) + 2 \sin(-45^\circ) = \sqrt{2} - \sqrt{2} = \mathbf{0}$
- $\theta = 0^\circ \implies \rho = 2 \cos(0^\circ) + 2 \sin(0^\circ) = 2(1) + 0 = 2$
- $\theta = 45^\circ \implies \rho = 2 \cos(45^\circ) + 2 \sin(45^\circ) = \sqrt{2} + \sqrt{2} = 2\sqrt{2} \approx 2.828$
- $\theta = 90^\circ \implies \rho = 2 \cos(90^\circ) + 2 \sin(90^\circ) = 0 + 2(1) = 2$

θ	-45°	0°	45°	90°
ρ for $P_1(1, 1)$	0	1.0	1.414	1.0
ρ for $P_2(2, 2)$	0	2.0	2.828	2.0

The parameter curves intersect at $(\rho_0, \theta_0) = (0, -45^\circ)$. Substituting back into $x \cos(-45^\circ) + y \sin(-45^\circ) = 0 \implies \frac{\sqrt{2}}{2}x - \frac{\sqrt{2}}{2}y = 0 \implies y = x$, recovering the line passing through both points.